

Proposal for the ECGI Challenge

Localization from relative geodesic comparisons between pacing sites

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Background. The goal is to localize an unseen ventricular activation origin from torso electrical measurements. Conventional ECGI reconstructs cardiac electrical activity on the heart surface and finds the source from the reconstructed activation map. A simpler pace-mapping approach [1] assigns the unknown signal to its most similar measured pacing template, which restricts the prediction to a previously measured location. The proposed method also uses recordings from known pacing sites, but instead of choosing the closest match, it combines their relative anatomical information to estimate a source anywhere on the ventricular surface, including locations that were never paced directly.

Related work. Pace mapping compares the unknown beat with recordings from known pacing sites and assigns it to the single measured site whose ECG is most similar. There is also mathematical work on locating an unknown point from comparisons such as “closer to A than to B ” [2]. I use this type of comparison to extend pace mapping beyond the measured sites and estimate the source directly on the ventricular surface.

Tested method. Instead of predicting a coordinate directly, I compare the unknown signal with pairs of measured pacing sites. For each pair A, B , the model estimates whether the source is geodesically closer to A or to B . Each comparison adds information about where the source could be, and combining many comparisons gives a score for every point on the ventricular surface.

Main idea. *Convert learned relative geodesic comparisons between known pacing sites into a localization estimate directly on the ventricular surface.*

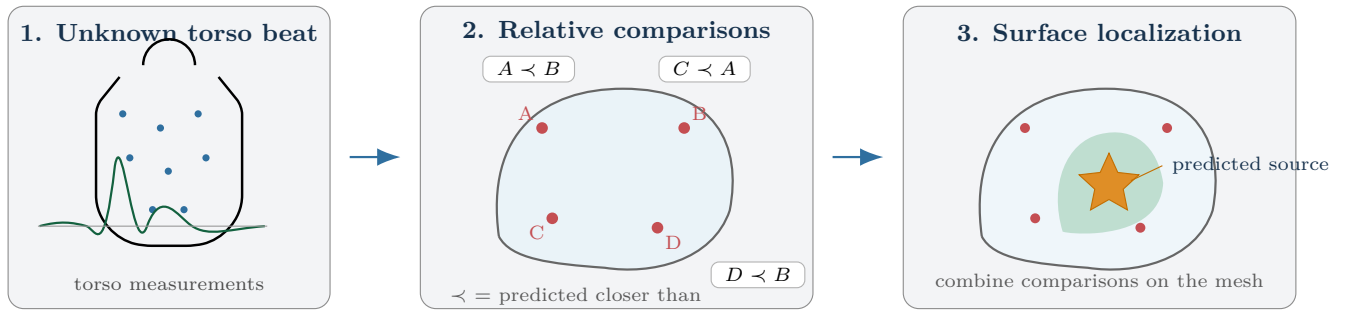


Figure 1: **Concept of the proposed method.** An unseen torso beat is converted into relative geodesic comparisons between known pacing sites. Combining these comparisons directly on the ventricular surface produces a source estimate without reconstructing a complete activation map.

Evaluation. For Pig2, I used 194 individual paced beats from 14 pacing sites on a 239-vertex epicardial sock mesh. I hid one complete pacing site at a time. The hidden site was not used to build templates, train the models, or choose their settings. Each pacing site counted once in the table, and error was measured as distance along the sock surface.

I compared three main approaches:

- **Nearest template:** choose the known pacing site whose torso signal looks most similar to the unknown beat.
- **Inverse reconstruction:** use torso recordings together with recordings taken from the heart surface at the known sites, reconstruct activity on the heart, and then choose the earliest activation.
- **Relative comparisons:** estimate whether the unknown source is closer to one known pacing site or another, then combine many such comparisons on the ventricular surface.

I tested four ways of combining the relative comparisons: using the probability values, using only their order, using both, or searching only near the closest template.

The median describes the error for a typical pacing site, while the “90% below” column shows the size of the larger errors.

Results. The nearest-template method had a median error of 42.7 mm.

Using the order of the relative-comparison scores reduced the median error to 31.6 mm and gave a better result on 10 of the 14 pacing sites. Using both the score order and the probability values gave the lowest median error, 28.9 mm.

Method	Mean mm	Median mm	90% below mm	Better than nearest template
Nearest template	42.8	42.7	63.6	–
Inverse reconstruction	30.5	31.1	45.0	9 of 14
Relative comparisons using probabilities	38.0	37.1	69.9	8 of 14
Relative comparisons using score order	36.1	31.6	79.5	10 of 14
Relative comparisons using both	37.0	28.9	69.9	8 of 14
Score order near the closest template	38.8	40.6	62.8	9 of 14

The inverse reconstruction had a similar median error of 31.1 mm and performed better than nearest-template matching on 9 of the 14 pacing sites. It also made fewer large errors. Unlike the proposed method, it used recordings taken from the heart surface at the known pacing sites during training.

The current Pig2 result shows that relative comparisons can improve over nearest-template matching and approach inverse reconstruction for a typical pacing site, although the inverse method is more consistent.

Conclusion. This is a promising first result from one animal. The next step is to keep the method unchanged and test it on the other animals.

References

- [1] A. SippensGroenewegen, H. Spekhorst, N. M. van Hemel, et al., “Localization of the site of origin of postinfarction ventricular tachycardia by endocardial pace mapping: body surface mapping compared with the 12-lead electrocardiogram,” *Circulation*, vol. 88, no. 5, pp. 2290–2306, 1993.
- [2] A. K. Massimino and M. A. Davenport, “As You Like It: Localization via Paired Comparisons,” *Journal of Machine Learning Research*, vol. 22, no. 186, pp. 1–39, 2021.